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ORIGINAL PAPER

Assessing glacial lake outburst flood risk in the Eastern Himalayas: a Bayesian neural network framework

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Abstract

Glacial lake outburst floods (GLOFs) represent a growing global hazard with increasing impacts in high-mountain regions across Asia, South America, and the Arctic. In the Eastern Himalayas, the frequency and intensity of GLOFs have risen markedly over the past three decades, driven by accelerated glacier retreat, altered precipitation patterns, and unregulated development in downstream valleys. However, systematic risk assessment remains limited, constrained by insufficient understanding of glacier–lake interactions and the lack of predictive models that integrate hazard complexity and uncertainty. These deficiencies weaken the effectiveness of early warning systems and adaptation strategies, not only in the Himalayas but also in other vulnerable mountain regions worldwide. This study presents a robust, two-stage methodology to assess GLOF risk in the Eastern Himalayas, offering a scalable framework relevant to other glaciated regions. In the first stage, we employ a multi-criteria decision-making (MCDM) approach integrating 14 parameters—including topographic configuration, geomorphology, hazard proximity, meteorological extremes, and anthropogenic exposure—weighted using a hybrid of Shannon Entropy and the Analytic Hierarchy Process (AHP). This model classifies glacial lakes into four GLOF risk categories: low, moderate, high, and very high. In the second stage, we implement a Bayesian Neural Network (BNN) to replicate and refine the initial classification while quantifying predictive uncertainty—marking the first application of BNNs in GLOF risk science. The model achieves an overall accuracy of 93.4%, identifying 66 low-risk, 74 moderate-risk, 56 high-risk, and 18 very high-risk lakes. Importantly, 87.77% of lakes exhibit predictive variance below 0.1, and none exceed 0.165, indicating high model reliability across risk categories. By integrating expert-driven decision-making with uncertainty-aware machine learning, this research offers a globally relevant advancement in GLOF hazard assessment. The methodology provides a transferable template for other climate-sensitive mountain regions facing similar threats. It supports the development of more adaptive, evidence-based disaster risk reduction strategies aligned with global frameworks such as the Sendai Framework for Disaster Risk Reduction and the UN Sustainable Development Goal. In an era of accelerating cryosphere change, such predictive and scalable approaches are critical for safeguarding downstream communities, infrastructure, and transboundary water security.

Extended author information available on the last page of the article

Keywords Earthquake · GLOF · Landslides · Risk assessment · AHP · Shannon entropy

1 Introduction

Despite growing concern over glacial lake outburst floods (GLOFs) in the Eastern Himalayas, significant gaps remain in the comprehensive risk assessment of potentially dangerous glacial lakes (PDGLs). Existing approaches are often constrained by limited parameter selection, reliance on deterministic threshold-based classification, exclusion of multiple hazard triggers such as seismicity and landslides, and the inability to quantify predictive uncertainty (Kougkoulos et al. 2018; McKillop and Clague 2007; Rounce et al. 2016; Zhang et al. 2022a, b). Most notably, there is a lack of unified models that can assess risk across all lake types—moraine-, ice-, and bedrock-dammed—while also integrating non-linear interdependencies among diverse contributing factors. These deficiencies pose critical challenges to disaster preparedness, especially in the data-sparse and rapidly evolving environments of the high Himalayas.

To address these limitations, this study proposes a Bayesian Neural Network (BNN)-based approach for GLOF risk assessment. The model is trained using risk labels generated through Fuzzy C-Means (FCM) clustering of TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) scores, which are themselves derived from hybrid Analytic Hierarchy Process (AHP) and Shannon Entropy weights. By combining expert judgment and objective information theory, this hybrid weighting scheme enhances the robustness of the multi-criteria decision-making (MCDM) process. The BNN model offers a key advancement over conventional deterministic methods by providing probabilistic predictions and uncertainty quantification, thereby supporting more informed decision-making in the context of hazard mitigation.

Glaciers in high-altitude environments serve as vital freshwater reservoirs, regulating streamflow and supporting downstream ecosystems and human settlements (Ageta et al. 2000). However, the accelerated retreat of glaciers due to anthropogenic climate change has led to the proliferation of glacial lakes, especially in the Eastern Himalayan region encompassing India, Bhutan, Nepal, and Tibet (Ahmed et al. 2021; Nie et al. 2013, 2021; Song et al. 2016; Veettil et al. 2016; Wang et al. 2015; Zhao et al. 2018). These lakes, often impounded by unstable moraines, are highly susceptible to sudden breaching, resulting in catastrophic GLOF events. The Eastern Himalayas have emerged as a hotspot, exhibiting three times higher GLOF susceptibility than other segments of the Himalayan belt (Bajracharya et al. 2014; Bajracharya and Mool 2009; Bolch et al. 2011; Rafiq et al. 2019).

Historical GLOFs, including those from Gaya Glacial Lake (Bhutan), Chorabari Lake (India), and Imja Tsho (Nepal), have caused extensive loss of life and damage to infrastructure, highlighting the urgent need for advanced risk assessment frameworks (Dubey and Goyal 2020; Komori et al. 2012; Mehta et al. 2012; Somos-Valenzuela et al. 2015; Wangchuk and Tsubaki 2024). GLOF triggers are diverse and often interdependent, including glacial retreat, avalanches, slope failures, earthquakes, and extreme precipitation (Avouac 2003; Carey 2005; England and Bilham 2015; Jain et al. 2013; Yongping 2004). For instance, sustained atmospheric warming destabilizes slopes and increases landslide risk, which, when combined with weakened moraine dams, significantly heightens the potential for outburst. These risks are compounded in tectonically active regions with steep terrain

and interconnected lake systems (Aggarwal et al. 2017; Allen et al. 2019; Veh et al. 2020; Zheng et al. 2021).

Recent events in Sikkim and Arunachal Pradesh underscore this compounded vulnerability. In October 2023, South Lhonak Lake in Sikkim, one of the region's most rapidly expanding glacial lakes, burst following intense rainfall and probable avalanche-induced overtopping, leading to the destruction of the Teesta III dam (Sattar et al. 2021; Sharma et al. 2018). Similarly, the 2024 GLOF from Sangnga Nehgu Lake in Arunachal Pradesh obliterated key pastoral routes vital to the Brokpa herding community, demonstrating the far-reaching socio-economic implications of these events.

In addition to climate-induced hazards, seismic events also exacerbate GLOF susceptibility. The 2015 Gorkha Earthquake in Nepal destabilized several moraine-dammed lakes, such as Tsho Rolpa and Barun Tsho, by inducing cracks in moraine structures and triggering landslides (Rafiq et al. 2019). Consequently, GLOF risk is shaped by a complex interaction of physical geography, meteorological extremes, anthropogenic exposure, and the legacy of past hazards.

The remoteness and inaccessibility of many Himalayan lakes hinder continuous field-based monitoring, reinforcing the value of remote sensing and geospatial modelling for GLOF risk assessment. However, current remote-sensing-based models exhibit several limitations. McKillop and Clague (2007) developed a remote-sensing approach for moraine-dammed lakes in British Columbia, yet it was constrained by its empirical structure and omission of downstream vulnerabilities. Rounce et al. (2016) introduced a heuristic rule-based model using global datasets, but it excluded seismic factors and was applicable only to moraine-dammed lakes. Other models, such as Zhang et al. 2022a, b; Kougkoulos et al. (2018), either ignored crucial factors like seismic activity or relied on equal weighting and threshold-based classification, failing to capture non-linear interactions.

Advancements in MCDM frameworks–particularly AHP and entropy-based weighting, have improved model reliability. AHP offers a structured method to integrate expert judgment (Rounce et al. 2016, 2017; Sharma et al. 2018), while Shannon Entropy provides objective weights based on the variability of input data (Arora et al. 2021). Hybrid methods have demonstrated better robustness; for instance, Vashistha et al. (2025) employed an AHP–Entropy model for GLOF risk in the Eastern Himalaya. While this framework effectively integrated expert judgement with data-driven weighting, it remained deterministic in nature, lacking adaptability to changing environmental conditions and the ability to quantify predictive uncertainty. Moreover, certain critical exposure and hazard factors were not explicitly addressed–such as glacier proximity, neighbouring lakes that can trigger cascading outburst events, and the influence of climate-driven extremes. These omissions limit the comprehensiveness of deterministic hybrid models and underscore the need for probabilistic approaches that can explicitly capture uncertainty while accounting for dynamic hazard–exposure interactions.

Modern engineering systems are witnessing a major transformation, as conventional first-principle models are now being supplemented with data-driven techniques. The convergence of big data, artificial intelligence, and machine learning has accelerated this shift, gaining significant traction across science and engineering domains (Anilkumar et al. 2023; Kalimullah et al. 2023, 2025; Tang et al. 2024). The growing interest in machine learning (ML) for hazard prediction has prompted efforts to integrate expert-driven frameworks with learning-based models. For example, Mondal et al. (2025) employed Artificial Neural

Networks (ANNs) trained on AHP-derived inputs. Yet, traditional ANNs remain deterministic and do not provide explicit uncertainty estimates–an essential requirement for disaster risk reduction in data-scarce regions. In contrast, BNNs provide probabilistic modelling capabilities and quantify epistemic uncertainty, offering a major improvement over existing methods (Guillet and Bolch 2023; Jospin et al. 2022).

This study introduces an integrated GLOF risk assessment framework that addresses these limitations. The key contributions include:

Comprehensive parameter inclusion Incorporating diverse risk factors including lake geometry, topography, geomorphology, exposure, meteorological variables, and proximity to past hazard events across the Eastern Himalayas.

Risk estimation via BNN A Bayesian Neural Network is trained on FCM-clustered risk labels derived from hybrid AHP-Shannon Entropy-weighted TOPSIS scores, enabling nuanced learning of parameter interdependencies.

Uncertainty quantification Providing probabilistic risk outputs to inform more robust decision-making under uncertainty.

Focused risk prediction for five PDGLs The model is applied to five well-studied, high-risk glacial lakes—Imja Tsho, Tsho Rolpa, Barun Tsho, Luggye Tsho, and South Lhonak—offering regionally and globally relevant insights.

Accordingly, this study proposes a machine-learning-based, data-informed, and uncertainty-aware framework for the automated risk assessment of glacial lakes larger than 0.01 km^2, suitable across all dam types. This contribution aims to support science-based prioritization, risk communication, and mitigation planning in one of the world's most climate-sensitive mountain systems.

2 Study area and materials required

2.1 Study area

The Eastern Himalaya is a geologically active and topographically diverse region, extending from approximately 21.63°N to 31.36°N latitude and 85°E to 98.73°E longitude. It spans northeastern India, Bhutan, Nepal, the southeastern Tibetan Plateau, and northern Myanmar, covering an estimated 147.12 million hectares. The topography map shows that the elevation of the study area ranges from − 117 m in the southern lowlands to 8781 m in the northern highlands, including Kangchenjunga (8586 m), the world's third-highest peak (Fig. 1). The region represents a critical transitional zone between the high-altitude Tibetan Plateau and the tropical lowlands of Southeast Asia. It is characterized by a humid, monsoon-driven climate, with substantial precipitation during the southwest monsoon (June–September).

Unlike the Western Himalaya glaciers (large and continuous), the Eastern Himalaya glaciers are generally smaller, more fragmented, and dispersed across steep terrain. Around 35–40% of the area remains snow-covered even during summer months (Krishna 2005),

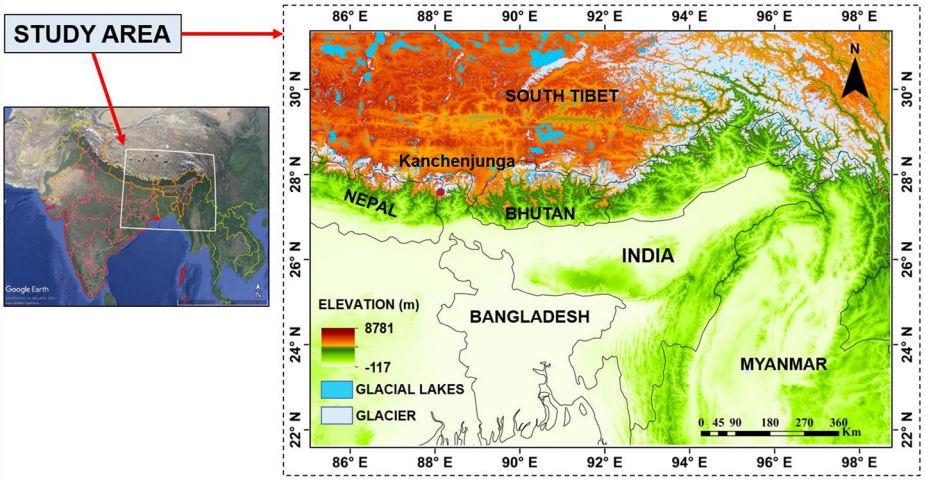


Fig. 1 Topography of the Eastern Himalayas study area. The inset map (left) shows the regional location within South Asia, while the main map (right) highlights Nepal, Bhutan, and adjoining parts of India, Bangladesh, Myanmar, and South Tibet. Glaciers and glacial lakes are concentrated in the high-elevation Himalayan belt, where the risk of glacial lake outburst floods (GLOFs) is most pronounced

supporting numerous small- to medium-sized glaciers. These glaciers are predominantly debris-covered and highly sensitive to temperature variations. Climatically, the Eastern Himalayas are undergoing significant and disproportionate warming, with an observed increase in annual mean temperature of approximately 0.01 °C per year (Li et al. 2016). This climatic trend is accelerating glacial retreat and contributing to the rapid formation and expansion of glacial lakes. This morphological characteristic, combined with ongoing climate-induced changes, has led to increased vulnerability to glacial lake outburst floods (GLOFs). The region is hydrologically significant, as major rivers such as the Brahmaputra, Teesta, and Manas originate from these glacier-fed systems and support agriculture, hydropower, and water security in downstream areas across India, Bhutan, Bangladesh, and China.

2.2 Parameters used in GLOF risk assessment

The GLOF risk assessment is performed based on 6 key factors, namely geomorphology, physical and geometrical characteristics of the lake, topography, spatial proximity to past natural hazards, meteorological conditions, and anthropogenic exposure. The geomorphology of the region is incorporated by the proximity to the neighbouring lakes. Further, the physical and geometrical characteristic of the lake factor is subdivided into 2 parameters as follows: (1) the area of the lake and (2) proximity to the glacier. Similarly, the risk associated with topography variation is analysed based on 2 sub-factors: (1) Elevation of the lake and (2) slope. Spatial proximity to past natural hazards is subdivided into 5 factors such as (1) proximity to landslides, (2) proximity to earthquakes, (3) magnitude of nearest earthquakes, (4) depth of the nearest earthquake, and (5) proximity to GLOF lakes. Next, meteorological conditions are divided into 2 sub-factors, (1) precipitation seasonality and (2) temperature seasonality. Anthropogenic exposure is subdivided into 2 factors, (1) Land

use land cover, and (2) Proximity to hydropower station. The aforementioned factors lead to the selection of 14 key parameters for GLOF risk assessment. Importantly, these parameters are decided based on free accessibility to data, non-recurring influence, and relevance in literature to update the risk model.

2.2.1 Presence of neighbouring glacial lakes

The presence of neighbouring glacial lakes, either upstream or downstream, is a critical factor in glacial lake outburst flood (GLOF) hazard assessment, as it can trigger cascading lake failures and substantially amplify flood impacts (Ahmed et al. 2022; Islam and Patel 2022). A notable example is the Lemthang Tsho GLOF that occurred on 28 June 2015, which was initiated by the collapse of a steep ice wall from a series of interconnected supraglacial ponds located upstream of Lemthang Tsho (Gurung et al. 2017). This collapse activated a subglacial drainage system, allowing water from the supraglacial ponds to flow into Lemthang Tsho through an englacial conduit. The resulting inflow caused a rise in lake level and the buildup of hydrostatic pressure, ultimately leading to the breach of the moraine dam, exacerbated by erosion and the dislodgement of a large boulder. The outburst released approximately 0.37 million cubic meters of water, with an estimated peak discharge ranging from 1253 to 1562 m^3/s, and affected infrastructure up to 30 km downstream. Similarly, the Tshojo GLOF of 29 April 2009 was triggered by the breach of supraglacial lakes on the Tshojo Glacier in the Pho Chu basin, resulting in downstream flooding (Gurung et al. 2017; Komori et al. 2012). Another significant example is the Zhangzangbo GLOF event of 1981, which was initiated by an ice-rock avalanche that triggered the outburst of the Zhangzangbo glacial lake, releasing a flood with a peak discharge of approximately 15,920 m^3/s. As the flood wave travelled downstream, it overtopped or breached the moraine dam of a secondary lake, Yarong Tsho, thereby amplifying the flood volume and its destructive impact in Nepal's Sun Koshi Valley (Xu 1988). These events highlight the necessity of considering neighbouring lakes in GLOF risk assessments, as their spatial proximity and hydrological connectivity can significantly affect lake stability and the overall intensity of a GLOF event. Therefore, the present study incorporates the distance between a glacial lake and its neighbouring lakes using ArcGIS, based on the rationale that a shorter inter-lake distance indicates a higher potential risk due to the increased likelihood of cascading failures. In this study, glacial lakes with an area greater than 0.1 km^2 (Wang et al. 2015) and located at an elevation greater than or equal to 2450 m are assessed for risk assessment. The elevation threshold was determined from Zhang et al. (2022a, b) study, in which 2451 m is identified as the minimum elevation of potentially dangerous glacial lakes. Accordingly, the dataset comprises 1144 glacial lakes.

2.2.2 Area of the lake

The area of the lake is closely linked to its hydrodynamic behaviour. Larger lakes can store more water, increasing the peak discharge during an outburst. Moreover, a larger area indicates weak dam boundaries, increasing the risk of dam erosion and debris flow during a GLOF. In this study, the HydroLAKES v1.0 dataset (Messager et al. 2016) is utilized to extract the surface area of lakes for GLOF risk evaluation.

2.2.3 Proximity to glacier

Lakes that are located near the glacier terminus are often more dynamic and hazardous due to their direct interaction with active glacial melt, ice calving, and supraglacial water drainage. These pro-glacial lakes tend to expand more rapidly as the glacier retreats, increasing hydrostatic pressure on moraine dams. They are also more vulnerable to sudden inputs such as icefalls or subglacial discharge, which can initiate overtopping waves or destabilize the dam structure. In contrast, lakes farther from the glacier are generally older, potentially more stable, and less influenced by active ice processes. However, they can still present significant hazards depending on factors such as intense precipitation, inflow from neighbouring lakes, or seismic triggers. In this study, the distance between each lake and its parent glacier is measured from the outer edge of the lake boundary by using the Near tool. Accordingly, glacier extents are delineated by applying the Normalised Difference Snow Index (NDSI) on cloud-free Sentinel-2 imagery, processed within the Google Earth Engine environment. An elevation threshold of 3500 m was applied using Shuttle Radar Topography Mission (SRTM) 30 m DEM, to refine glacier detection and minimize false positives from seasonal snow or non-glaciated high-elevation areas.

2.2.4 Elevation and slope

Elevation and slope are important topography indicators of a glacial lake's potential to trigger a GLOF. High-elevation lakes have greater potential energy. In the event of dam failure, this energy can translate into high-velocity flood waves with increased erosive power, thereby amplifying the downstream impact. Further, steeper slopes near glacial lakes can promote rapid downstream movement of floodwaters, increasing their erosive power and destructive impact. For instance, Govindha Raj (2010) found that glacial lakes with downstream slopes greater than 10° are potentially dangerous. Che et al. (2014) suggested that a steep slope (> 8° between lake and glacier increases the risk of calving and outburst. Therefore, the present study uses the steepest slope within a 500 m buffer zone surrounding each lake for risk evaluation, reflecting the terrain's potential to initiate or amplify GLOF events. The present study evaluates the elevation and slope by using SRTM 30 m DEM through ArcGIS.

2.2.5 Proximity to landslide

Landslides, often triggered by excessive precipitation or seismic activity, can act as direct triggers for GLOFs when they fall into lakes, causing a sudden displacement of water and a massive amount of debris flow. For instance, in the 1985 Dig Tsho GLOF in Nepal, a massive landslide was a contributing factor in initiating the breach of the moraine dam (Vuichard and Zimmermann 1987). Thus, this study evaluates the distance of glacial lakes from landslides using the Near tool. Figure 2 presents the comprehensive view of landslides, glaciers and glacial lakes in the study area, evidencing potential risk of lake outburst. In this study, Kirschbaum et al. (2015) and Kirschbaum et al. (2010) research was used to extract the landslide dataset.

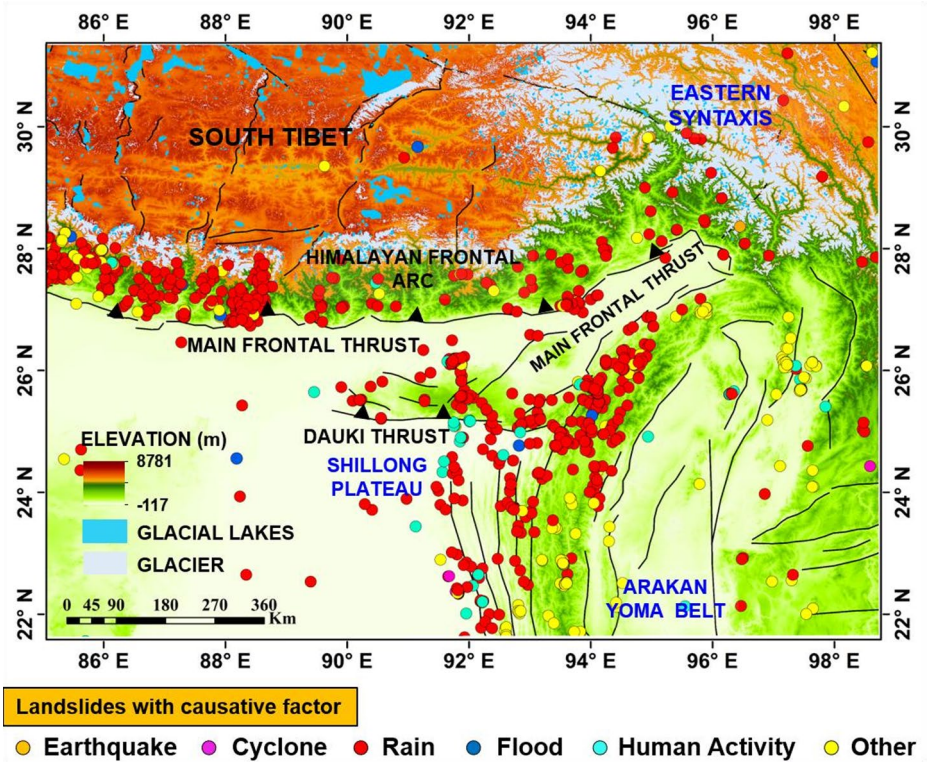


Fig. 2 Spatial distribution of landslides in the Eastern Himalayas categorized by causative factors, overlaid on an elevation map derived from SRTM 30 m DEM. The map also displays lake and glacier extents along with major tectonic structures. Coloured circles represent landslide events triggered by rainfall, earthquakes, cyclones, floods, human activity, and other miscellaneous causes

2.2.6 Proximity to earthquake

The Eastern Himalaya encompasses five of the most seismically active regions: the Eastern Syntaxis, the Arakan Yoma Belt, the Shillong Plateau (SP), the Himalayan Frontal Arc (HFA), and South Tibet. Earthquakes in these regions are predominantly of shallow depth, with intermediate magnitudes and spatially variable frequencies. Between 1964 and 2020, the Eastern Syntaxis experienced 247 recorded earthquakes, the Arakan Yoma Belt 727 earthquakes, the Shillong Plateau 83 earthquakes, the HFA 166 earthquakes, and South Tibet 124 earthquakes (Tiwari and Paudyal 2022). The prevalence of shallow, moderate-magnitude earthquakes poses a significant threat to glacial lake stability, as seismic shaking can trigger moraine dam failures or landslides, leading to GLOF events, as observed from the 2015 Gorkha earthquake (Cook et al. 2018). Recognizing this risk, the present study incorporates not only the proximity of lakes to the nearest earthquake epicentres but also accounts for the magnitude and focal depth of these events in the GLOF hazard assessment. For this purpose, updated earthquake data from 1960 to 2024 were sourced from the United States Geological Survey (USGS) database to ensure a comprehensive evaluation. Accordingly, the proximity of glacial lakes to landslides is assessed using the Near tool in

ArcGIS. Figure 3 This section presents the spatial distribution of earthquakes with their magnitude and depth variation. Additionally, active faults are shown to show the criticality of the region.

2.2.7 Distance of lake from past GLOF events

Past GLOF events are considered in proximity analysis to identify lakes that lie in historically affected regions or downstream of former outburst paths, where recurrence risk is significantly elevated. Lakes situated close to past GLOF locations are more likely to exhibit similar topographic, geomorphic, and climatic settings that predispose them to future outbursts. The spatial distribution of past GLOF events (Fig. 4) was derived from the studies by Lützow et al. (2023) and Zhang et al. (2022a, b), which together documented 75 GLOF occurrences within the defined study region.

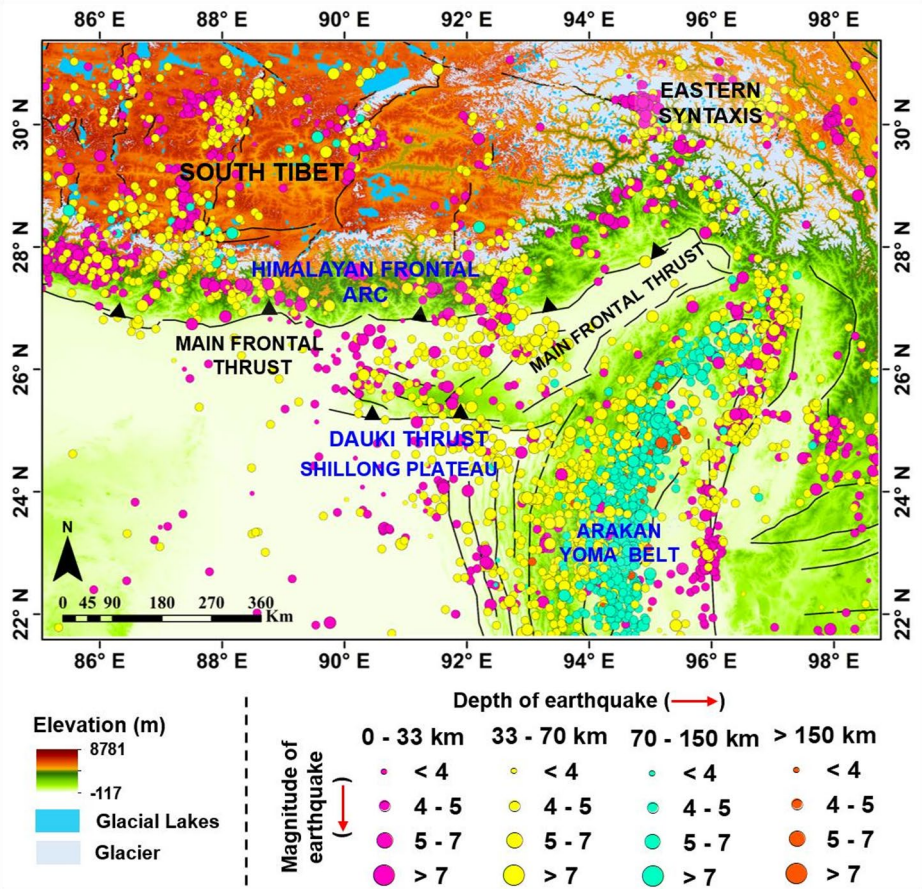


Fig. 3 Spatial representation of earthquakes with their magnitude and depth variation over the Eastern Himalaya.
(*Source* USGS). Active faults are also presented and obtained from Taylor and Yin (2009).

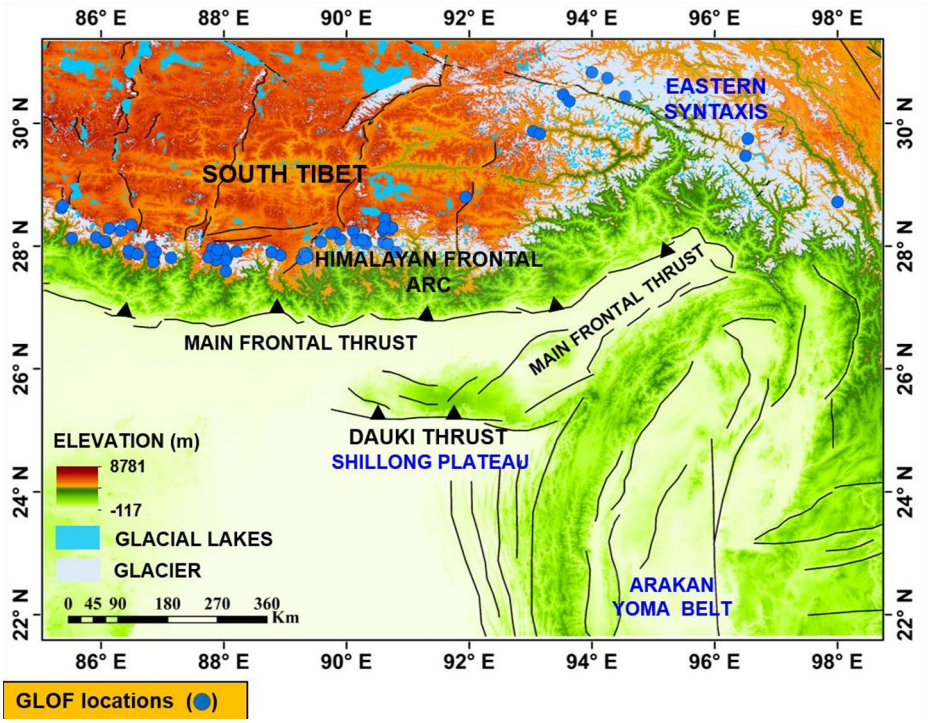


Fig. 4 Spatial distribution of recorded GLOF (blue circle) events over the Eastern Himalayas

2.2.8 Precipitation and temperature seasonality

Precipitation and Temperature are key climate change parameters of GLOFs, as increased temperature accelerates glacier melt and ice calving, while intense or variable precipitation can rapidly increase lake volume. Both parameters have the potential to increase mass inflow into glacial lakes, thereby elevating the risk of outburst events. This study considers two key BioClim variables, particularly BioClim 4 and BioClim 15 for long-term climatic assessment in GLOF-prone regions. BioClim 4, or Temperature Seasonality, represents the standard deviation of monthly mean temperatures and effectively quantifies the degree of temperature fluctuation throughout the year. Higher values of BioClim 4 indicate greater seasonal temperature variability, which can influence glacier stability through repeated freeze-thaw cycles and variable melt rates. BioClim 15, or Precipitation Seasonality, is the coefficient of variation of monthly precipitation totals. This variable captures the extent to which precipitation is unevenly distributed across the year. Higher values suggest more irregular or seasonally concentrated precipitation, which may contribute to the rapid accumulation of lake water, increasing the hydrostatic pressure on moraine dams and the likelihood of a GLOF event. Detailed descriptions of these variables can be found in the study by O'Donnel and Ignizio (2012).

2.2.9 Land use land cover (LULC) pattern

The type of land cover surrounding a glacial lake plays a key role in influencing the local hydrological behaviour, particularly in terms of how water flows, accelerated and resisted in the terrain. In this study, Sentinel-2 LULC data at 10-meter resolution was used to identify the majority LULC class surrounding the glacial lake (buffer distance = 500 m), where LULC class may vary as glacier, water, trees, crops, flooded vegetation, rangeland, and built area, providing detailed insight for regional-scale analysis. Further, this qualitative data was converted into a quantitative form by ordering the given classes into an increasing risk order, which varies from 1 to 7, with 1 indicating the lowest risk LULC class and 7 indicating the maximum risk LULC class.

Water bodies (7) pose the highest risk by expanding flood volumes, while built-up areas (6) are highly vulnerable due to infrastructure and population exposure. Rangelands (5) lack dense vegetation, leading to increased soil erosion and sediment transport into lakes, which can weaken natural dams. Flooded vegetation (4) offers temporary hydraulic resistance but can worsen debris flow if uprooted during a flood. Tree cover (3) reduces the flood velocity through canopy and root resistance, but uprooted trees may become hazardous debris. Croplands (2) generally reduce flow speed but mainly contribute to economic loss rather than structural risks. Ice-covered areas (1) are considered the least hazardous unless subjected to rapid melting, which may raise lake volumes. Figure 5 illustrates the LULC class and glacial lakes across the Eastern Himalayas, overlaid with major faults, glaciers, and hydropower stations, highlighting regions where LULC may influence glacial lake stability and associated hazards.

2.2.10 Proximity to a hydropower station

GLOF can cause catastrophic damage to turbines, dams, tunnels, and transmission systems if the flood reaches these installations. A hydropower station located downstream of a potentially unstable glacial lake faces high economic, structural, and human safety risks. Moreover, disruption of electricity supply can affect large populations and critical services. Therefore, assessing how close a lake is to such infrastructure helps prioritize lakes for monitoring, early warning systems, and mitigation planning. Figure 5 presents a comprehensive view of hydropower stations and waterbodies. The dataset for hydropower stations is obtained from Monitor (2024).

3 Methodology

The present study proposes a two-stage framework for Glacial Lake Outburst Flood (GLOF) risk assessment that synergistically integrates multi-criteria decision analysis (MCDA) with a probabilistic machine learning approach. In the first stage, a comprehensive set of 14 parameters influencing GLOF susceptibility was identified and categorized into positive and negative risk indicators. These indicators were normalized and assessed for multicollinearity using the Variance Inflation Factor (VIF). Following this verification, the relative importance of the indicators was quantified through a hybrid weighting approach that integrates the AHP and SE methods. The resulting hybrid weights were incorporated into a TOPSIS-

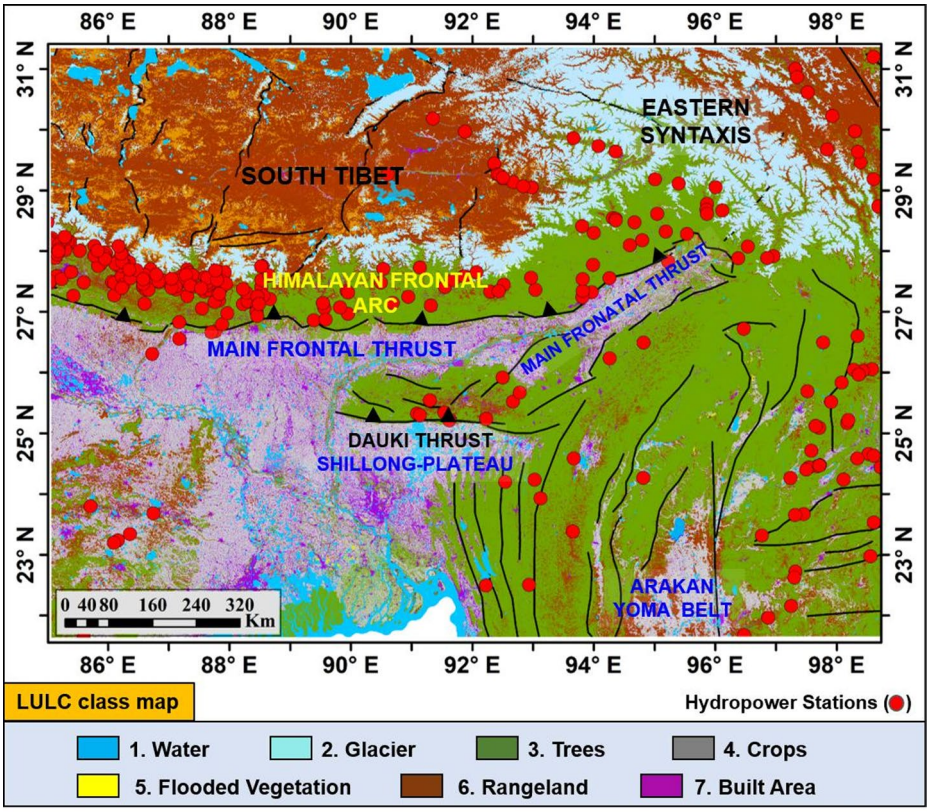


Fig. 5 LULC map with overlaid hydropower stations (red circle) and major faults

FCM (Technique for Order Preference by Similarity to Ideal Solution–Fuzzy C-Means) clustering framework to generate preliminary risk labels for each glacial lake, which subsequently served as training targets for the supervised model. In the second stage, a BNN was trained to learn the nonlinear and complex relationships between the 14 input features and the multi-class GLOF risk levels (very high, high, moderate, low). Bayesian optimization was employed to fine-tune the model's hyperparameters, thereby enhancing predictive performance over the validation set. The final BNN model thus offers a data-driven, uncertainty-aware classification tool for robust risk classification of glacial lakes across the Eastern Himalayas. A schematic flowchart of the proposed methodology is shown in Fig. 6.

3.1 Estimation of hybrid AHP-SE weights

Reliable estimation of parameter weights is paramount in MCDA-based GLOF risk modelling. This study employs a hybrid scheme that integrates subjective expert judgments via the AHP with objective data-driven variability captured through Shannon Entropy, thereby enhancing the robustness of the final weights.

The AHP is a widely recognized decision-support technique for solving multi-criteria problems by structuring them into a hierarchical form and quantifying the relative prefer-

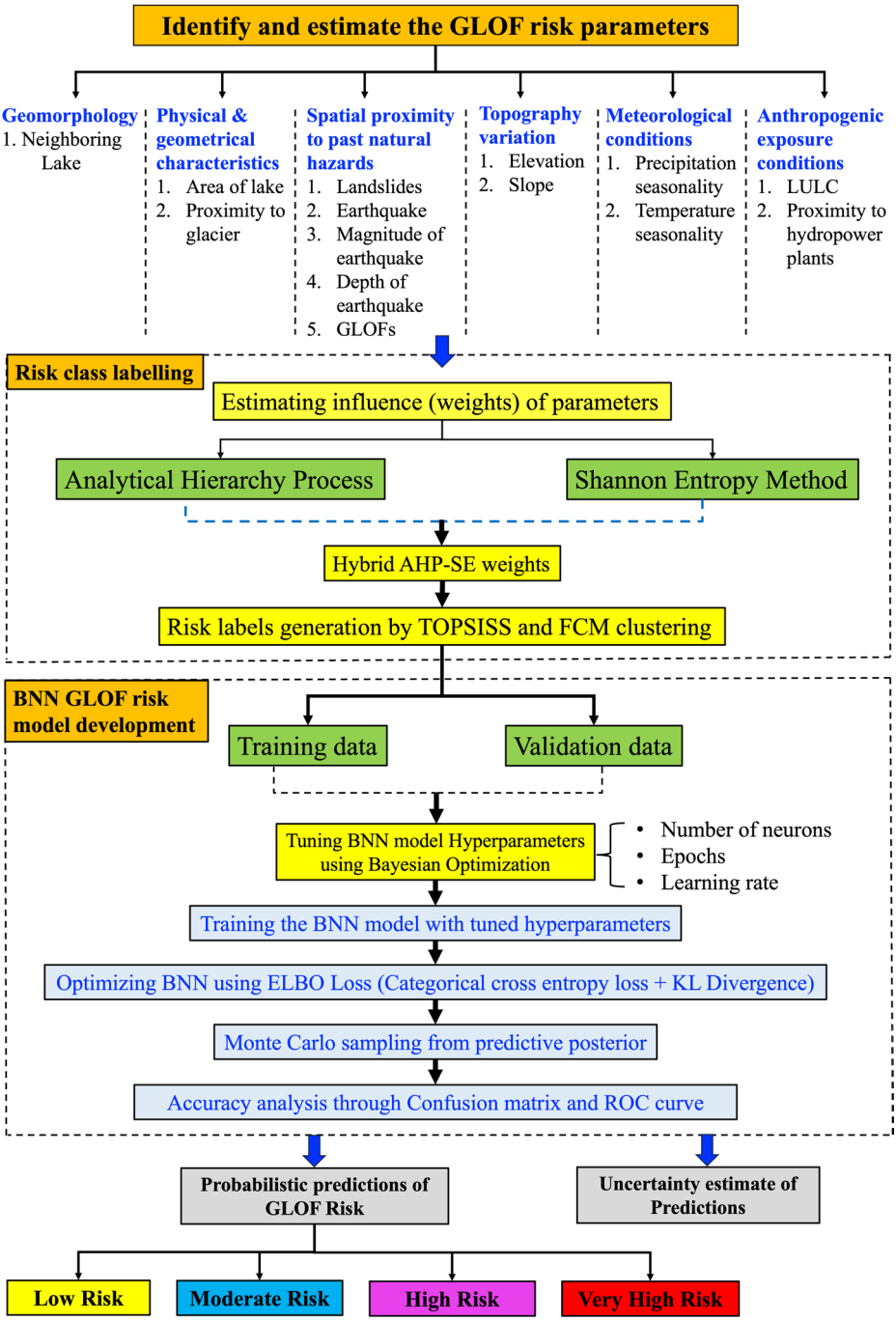


Fig. 6 Framework for probabilistic GLOF risk predictions using BNN model

ence of decision criteria based on expert opinions (Saaty 2008). To initiate the process, a pairwise comparison matrix (PCM) is constructed where each element a_{ij} denotes the importance of criterion *i* relative to criterion *j*. The judgments are based on a fundamental 9-point scale, where values of 1, 3, 5, 7, and 9 correspond to equal, moderate, strong, very strong, and extreme importance, respectively, while intermediate values (2, 4, 6, 8) capture nuanced preferences. The reciprocal condition is enforced, meaning $a_{ji} = 1/a_{ij}$ and all diagonal elements are unity $(a_{ii} = 1)$. After constructing the PCM, the matrix is normalized column-wise and the relative priority weights for each criterion are computed by averaging the normalized values across each row.

To assess the internal consistency of the expert judgments, the consistency index (CI) and consistency ratio (CR) are calculated. The maximum eigen value λ_{max} of the PCM is first derived, and the CI is obtained using the expression $\mathrm{CI} = (\lambda_{max} - n) / (n - 1)$, where n is the number of criteria. The CR is then computed as CR = CI / RI, where RI is the Random Index dependent on the matrix order and obtained from Saaty's table (Saaty 2008). A CR value below 0.10 indicates acceptable consistency which confirms that the resulting weights are logically reliable and suitable for use in the following stages of risk assessment.

In parallel, the SEw method is applied to estimate objective weights by evaluating the inherent informational diversity of each criterion across multiple alternatives. The entropy concept originates from information theory and quantifies the level of uncertainty associated with a distribution. For this purpose, a decision matrix is constructed where each entry denotes the performance of the *i* alternative with respect to the *j* criterion. To maintain consistent comparison across all criteria the raw data undergoes a normalization process. For positive indicators (where higher values imply greater risk), normalization is achieved using the formula $r_{ij} = (d_{ij} - min(d_j)) \ / \ \left(max(d_j) - min(d_j)\right)$. Conversely, for negative indicators (where lower values signify greater risk), the formula $r_{ij} = \left(max(d_j) - (d_{ij})\right) \ / \ \left(max(d_j) - min(d_j)\right)$ is used. After normalizing the decision matrix, entropy values E_j are calculated for each criterion by using the equation $E_j = -k \sum_{i=1}^{m} r_{ij} \ln r_{ij}$, where $k = 1/\ln(m)$ is a constant and m denotes the number of alternatives. Criteria with more variability (i.e., more informative) receive lower entropy scores and consequently higher weights, while uniform criteria yield higher entropy values and lower weights. The degree of diversification is then calculated as $d_j = 1 - E_j$, and the final entropy weight for the *j* criterion is derived by normalizing these values as $w_j = d_j / \sum_{j=1}^{n} d_j$.

Before integrating these weights with those derived from AHP, the indicators were assessed for multicollinearity using the Variance Inflation Factor (VIF). As shown in Table 1, the VIF values ranged from 0.803 to 3.801, all well below the critical threshold of 5, confirming that no significant multicollinearity exists and that each parameter contributes unique information.

To assess the agreement between the AHP and Shannon Entropy-derived rankings, Spearman's rank correlation coefficient (ρ) is used, which measures the strength of the monotonic relationship between the two ranked weight sets. A correlation value $\rho > 0.7$ indicates strong alignment, permitting direct combination of the weights, while $\rho \leq 0.4$, it reflects poor agreement and requires deeper investigation prior to integration. Next, if ρ varies as $0.4 < \rho \leq 0.7$ suggests moderate agreement, justifying cautious integration through a β combination. The combination coefficient (β) is optimized by minimizing the Kullback-

Table 1 Multicollinearity assessment of GLOF risk parameters by using VIF

Risk parameters	VIF
Neighbouring lake	0.804
Area	1.065
Proximity to nearest glacier	1.435
Elevation	2.420
Slope	1.173
Proximity to nearest landslides	3.801
Proximity to nearest earthquakes	1.137
Magnitude of earthquake	1.045
Depth of earthquake	0.963
Proximity to nearest GLOF	3.428
Precipitation seasonality	1.588
Temperature seasonality	1.473
Land cover	1.154
Proximity to nearest hydropower plant	2.330

Leibler (KL) divergence, which quantifies the dissimilarity between the two weight distributions. The resulting hybrid weights is obtained by using the equation as follows:

$$w_{j(AHP-SEw)} = \beta \;\; w_{j(AHP)} + (1 - \beta \;) \, w_{j(SEw)} \tag{1}$$

where $\beta \in [0,1]$, and $w_{j(AHP-SEw)}$ represent both the subjective importance derived from expert knowledge and the objective relevance inferred from data patterns.

To classify glacial lakes based on risk, this study integrates the TOPSIS ranking method with Fuzzy C-Means (FCM) clustering. TOPSIS evaluates each lake by comparing its weighted normalized parameters to an ideal (highest risk) and a negative-ideal (lowest risk) solution. The risk score is computed as the relative closeness to the ideal solution, where higher values indicate greater risk. These continuous risk scores are then grouped into four categories like very high, high, moderate, and low, using FCM clustering. Unlike hard classification, FCM assigns degrees of membership to each lake, allowing partial association with multiple risk levels. Cluster centers and membership values are updated iteratively until convergence, and final classification is based on the highest membership degree.

3.2 BNN for multi-class GLOF risk classification

In recent years, Glacial Lake Outburst Flood (GLOF) risk assessment has often relied on deterministic Multi-Criteria Decision-Making (MCDM) methods such as the AHP integrated with TOPSIS, or hybridized with objective weighting methods like entropy. While these frameworks provide a structured way to incorporate expert knowledge and multiple risk factors, they suffer from critical limitations. Specifically, these models are static in nature, unable to learn from new data or adjust their decision boundaries dynamically. Their reliance on pre-assigned weights and rule-based thresholds introduces subjectivity and rigidity, making them prone to inconsistencies when applied to new geographical regions or evolving climatic conditions. Even hybrid approaches like AHP-based ANNs, though data-driven, typically operate as black-box predictors with fixed architectures trained on point estimates, failing to offer meaningful insights into predictive uncertainty and lacking a Bayesian formulation to generalize beyond the training dataset.

In contrast, the developed BNN model represents a learnable and generalizable framework that can capture the complex, nonlinear interdependencies between geophysical, climatic, and topographic variables influencing GLOF risk. Unlike static models, BNNs continuously improve as new labelled data becomes available, enabling the model to adapt to diverse glacial environments across space and time. The learnability of BNNs ensures that they do not merely replicate rule-based outputs but infer patterns and relationships from data, offering superior generalization to unseen scenarios, a key requirement for anticipating emerging or undocumented glacial lake hazards.

3.3 Probabilistic modelling in BNN

For GLOF risk assessment, where predictions must be robust under data variability and modelling ambiguity, BNNs serve as an advanced method for classifying lakes into multiple risk categories like very high, high, moderate, and low-risk lakes. The proposed BNN model maps a 14-dimensional input vector $x \in \mathbb{R}^{14}$, comprising key physical characteristics, topography geomorphological, exposure, and climatic factors, to a probabilistic prediction over the risk classes represent the categorical output corresponding to the $y \in \{1,2,3,4\}$, risk level (very high, high, moderate, low). BNN models the posterior distribution over weights θ given data $M = \left\{(x_i, y_i)_{i=1}^{N}\right\}$ using Bayes' rule:

$$p(\boldsymbol{\theta}\,|\boldsymbol{M}) = \frac{p(M|\boldsymbol{\theta})\,p(\boldsymbol{\theta})}{p(M)} = \frac{\prod_{i=1}^{N} p\left(\mathscr{y}_i|x_i, \boldsymbol{\theta}\right) p(\boldsymbol{\theta})}{\int \prod_{i=1}^{N} p\left(\mathscr{y}_i|x_i, \boldsymbol{\theta}\right) p(\boldsymbol{\theta})\, d\boldsymbol{\theta}}, \tag{2}$$

where, N represents the total number of training samples. Given a trained posterior, the predictive distribution for a new input x^* becomes:

$$p\left(\mathscr{y}^*|x^*, \boldsymbol{\theta}\right) = \int_{\boldsymbol{\Theta}} p\left(\mathscr{y}^*|x^*, \boldsymbol{\theta}\right) p(\boldsymbol{\theta}\,|\boldsymbol{M})\, d\boldsymbol{\theta}\,. \tag{3}$$

The posterior is typically intractable due to the high-dimensional integral over the parameter space. To overcome intractability, stochastic variational inference (SVI) approximates the true posterior $p(\boldsymbol{\theta}\,|\boldsymbol{M})$ with a simpler distribution $q_{\psi}\ (\theta)$, parameterized by ψ, from a computationally tractable family Q. The variational parameters ψ are optimized by minimizing the KL divergence between the approximate and true posterior as:

$$\psi^{*} = \underset{\psi}{\operatorname{argmin}} \mathrm{KL}\left(q_{\boldsymbol{\psi}}\ (\boldsymbol{\theta})\ \|p\ (\boldsymbol{\theta}\mid M)\right) = \underset{\psi}{\operatorname{argmax}} \mathcal{L}(\boldsymbol{\psi}), \tag{4}$$

where, ψ^{*} is the optimal set of variational parameters. The above condition is equivalent to minimizing the evidence lower bound ($\mathcal{L}(\boldsymbol{\psi})$):

$$\mathcal{L}(\boldsymbol{\psi}) = -\mathbb{E}_{q_{\boldsymbol{\psi}}(\boldsymbol{\theta})}\left[\log p(M \mid \boldsymbol{\theta})\right] + \mathrm{KL}\left(q_{\boldsymbol{\psi}}\ (\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta})\right), \tag{5}$$

In practice, it becomes:

$$\mathcal{L}(\boldsymbol{\psi}) \approx\ -\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{N} \log p\left(y_i|\boldsymbol{x_i}\,,\ \boldsymbol{\theta_t}\right) + \mathrm{KL}\left(q_{\boldsymbol{\psi}}\ (\boldsymbol{\theta}) \,\|\, p(\boldsymbol{\theta})\right), \tag{6}$$

where, $\boldsymbol{\theta}_t \sim q_{\psi}(\boldsymbol{\theta})$ and T is the number of Monte Carlo samples.

For multi-class classification, the network's final layer outputs class (c) logits $z \in \mathbb{R}^4$, followed by a softmax transformation:

$$\mathrm{p}(\mathrm{y}=\mathrm{c} \mid \mathbf{x}, \boldsymbol{\theta}) = \frac{e^{zc}}{\sum_{j=1}^{4} e^{zj}}, \quad c = 1,2, \ldots, 4. \tag{7}$$

The softmax output yields normalized class probabilities, representing the model's estimated likelihood for each risk category. These probabilities are used during training to compute the categorical cross-entropy loss, which measures the divergence between the predicted probability distribution and the true class label. During inference, the same softmax outputs provide the predictive class probabilities for new inputs.

After optimization, predictions for a new input x^* are obtained by marginalizing over the learned approximate posterior:

$$\widehat{p}\left(y^* \mid x^*\right) = \int p(y^* \mid x^*, \boldsymbol{\theta}) q_{\psi^*}(\boldsymbol{\theta})\, d\boldsymbol{\theta} \approx \frac{1}{T}\sum_{t=1}^{T} p\left(y^* \mid x^*, \boldsymbol{\theta}_t\right), \tag{8}$$

where, $\boldsymbol{\theta}_t \sim q_{\psi^*}(\boldsymbol{\theta})$ and T is the number of Monte Carlo samples.

3.4 Hyperparameters optimization

BNN is characterized by a set of architectural and training hyperparameters that significantly influence its performance and generalization capability. These include the number of hidden layers, neurons per layer, activation functions, batch size, learning rate and number of epochs.

Optimizing hyperparameters is a critical step in neural network design, significantly affecting convergence, generalization, and predictive performance. Moreover, suboptimal choice of hyperparameters can lead to overfitting, underfitting, or slow and unstable convergence.

To overcome these limitations, this study adopts Bayesian Optimization (BO). BO is a model-based global optimization technique well-suited for black-box functions where derivative information is unavailable or evaluations are expensive, as is the case with BNN training (Wu et al. 2019). Formally, let X be the space of hyperparameter configurations and let $f : X \to \mathbb{R}$ represent the objective function (e.g., validation loss or negative ELBO). The BO approach can be presented as:

$$\boldsymbol{x}^* = \arg\min_{x \in X} f(\boldsymbol{x}) \tag{9}$$

where $\boldsymbol{x}^*$ represents the optimal set of hyperparameters minimizing the objective function.

BO builds a probabilistic surrogate model of the objective function, commonly a Gaussian Process (GP) or a Tree-structured Parzen Estimator (TPE), which is cheaper to evaluate and update after each observation. The surrogate model guides the selection of new hyperparameter configurations through an acquisition function (e.g., Expected Improvement or Upper Confidence Bound), which balances the exploration of uncertain regions with the exploitation of areas expected to yield better performance. This strategy improves search

efficiency by using prior evaluations to inform future sampling, avoiding the redundancy and inefficiency often associated with brute-force methods (grid search or random search).

3.5 Uncertainty analysis

Predictive uncertainty is a combination of aleatoric and epistemic uncertainty. Aleatoric uncertainty represents the noise inherent in data, while epistemic uncertainty captures uncertainty due to limited training data or lack of model confidence.

For softmax output $\boldsymbol{p}_t = p(\mathscr{y}^* \mid x^*, \theta_t)$, the predictive variance is as follows:

$$Var\ (\boldsymbol{y}^* \mid \boldsymbol{x}^*) = \underbrace{\frac{1}{T}\sum_{t=1}^{T}\left[\mathrm{diag}\,(\boldsymbol{p}_t) - \boldsymbol{p}_t\boldsymbol{p}_t^T\right]}_{\text{aleatoric}} + \underbrace{\frac{1}{T}\sum_{t=1}^{T}(\boldsymbol{p}_t - \bar{\boldsymbol{p}})(\boldsymbol{p}_t - \bar{\boldsymbol{p}})^{\boldsymbol{T}}}_{\text{epistemic}},\ \bar{\boldsymbol{p}} = \frac{1}{T}\sum_t \boldsymbol{p}_t \quad (10)$$

This combination enables risk stratification with confidence intervals in the GLOF, aiding proactive monitoring and response planning.

3.6 Application

After developing the BNN model, this study predicts the probabilistic GLOF risk over the 5 most frequently reviewed PDGLs, namely Imja Tsho, Tsho Rolpa, Barun Tsho, Luggye Tsho, and South Lhonak Lake. The selection of these lakes is based on their criticality, which is determined by a combination of factors such as rapid glacial retreat and lake expansion, the presence of unstable moraine dams, historical records of outburst events or strong indications of future risk, and their proximity to downstream populations, infrastructure, and hydropower facilities. These lakes are geographically distributed across Nepal, Bhutan, and India, representing different Himalayan sub-regions, allowing for a broader assessment of the model's applicability.

4 Results and discussion

4.1 Estimation of AHP-SE weights

In this study, the relative importance of fourteen parameters influencing GLOF susceptibility was determined using a hybrid approach that combines the AHP and SE methods. The AHP weights were derived through 91 pairwise comparisons over eight iterations, resulting in a consistency ratio (CR) of 0.074, which is below the acceptable threshold of 0.1 and confirms the reliability of expert judgments. Simultaneously, the SE method was applied to quantify the variability and information content of each parameter based on the distribution of input data, ensuring an objective estimation of weights without expert bias. Following the individual weight estimation, a consistency check was performed using Spearman's Rank Correlation Coefficient (ρ), which was found to be 0.5209. As the value of ρ falls within the range $0.4 < \rho \leq 0.7$, the weights from AHP and SE methods show moderate agreement and cannot be directly averaged. Therefore, a hybrid combination was adopted by applying a β-weighted strategy, where β equals 0.2625, to minimize the Kullback–Leibler (KL)

divergence between the two sets of weights. The final AHP-SE weights were obtained by combining 26.258% of the AHP weights and 73.742% of the SE weights, ensuring balanced representation of subjective expert input and objective data variability. Table 1 presents the AHP weights, SE weights, and the resulting AHP-SE hybrid weights along with their relative ranking.

As presented in Table 2, lake area emerged as the most influential parameter in GLOF risk assessment, with a combined AHP–SE weight of 0.2479, followed by elevation (W4 = 0.1892), land cover (W13 = 0.0871), earthquake magnitude (W8 = 0.0797), precipitation seasonality (W11 = 0.0688), and proximity to previous GLOF events (W10 = 0.0538). In contrast, several parameters exhibited relatively lower importance, with weights below 0.05. These include proximity to landslides (W6 = 0.0489), temperature seasonality (W12 = 0.0485), proximity to hydropower plants (W14 = 0.0434), proximity to seismic epicentres (W7 = 0.0368), presence of neighbouring lakes (W1 = 0.0310), proximity to glaciers (W3 = 0.0273), and earthquake depth (W9 = 0.0224). Among all, slope (W5 = 0.0111) received the lowest combined weight, indicating its comparatively limited influence on GLOF susceptibility in the study region. Overall, the results align with the physical understanding of GLOF mechanisms, supporting the effectiveness of the hybrid AHP–SE model in prioritising influential parameters.

After evaluating the hybrid AHP-SE weights, the labelled data of 4 classes (low, moderate, high, and very high risk) is generated for training the BNN model using TOPSIS and FCM clustering. The labelled data was selected after fifty-four iterations, following which there was a minimum improvement in FCM outcomes.

Table 2 Weight of parameters based on AHP, SE and AHP-SE method. Additionally, rank of parameters is presented using hybrid weighing (AHP-SE) approach

Risk parameters	AHP weight	SE weight	AHP-SE weight	Rank (AHP-SE)
Neighbouring lake (W1)	0.1064	0.0041	0.0310	11
Area (W2)	0.2136	0.2601	0.2479	1
Proximity to nearest glacier (W3)	0.0498	0.0193	0.0273	12
Elevation (W4)	0.1250	0.2121	0.1892	2
Slope (W5)	0.0247	0.0063	0.0111	14
Proximity to nearest landslides (W6)	0.0659	0.0429	0.0489	7
Proximity to nearest earthquakes (W7)	0.0563	0.0299	0.0368	10
Magnitude of earthquake (W8)	0.0572	0.0877	0.0797	4
Depth of earthquake (W9)	0.0434	0.0149	0.0224	13
Proximity to nearest GLOF (W10)	0.0540	0.0537	0.0538	6
Precipitation seasonality (W11)	0.0524	0.0746	0.0688	5
Temperature seasonality (W12)	0.0491	0.0484	0.0485	8
Land Cover (W13)	0.0666	0.0943	0.0871	3
Proximity to nearest hydropower plant (W14)	0.0352	0.0518	0.0434	9

4.2 BNN model simulation

The BNN was trained on a curated dataset comprising 14 input features extracted for each glacial lake. These features were standardized and mapped to discrete GLOF risk levels—low, moderate, high, and very high, using hard labels derived from initial fuzzy classifications. This dataset was randomly split into training and validation subsets using an 80 : 20 ratio to ensure stratified representation of all risk classes.

In the proposed BNN framework, each weight and bias parameter is modeled as a random variable following a Gaussian distribution. Specifically, the prior over the parameters is defined as a univariate normal distribution $\mathcal{N}(\mu\ ,\ \sigma^{\ 2})$, where μ is the mean and $\sigma^{\ 2}$ is the variance. In practice, the prior is set to a standard normal distribution with zero mean and unit variance. During training, the network learns an approximate posterior distribution for each parameter by optimizing the mean and variance of a variational Gaussian. To ensure optimal predictive performance and robust uncertainty quantification, BO was employed to tune the BNN's hyperparameters. BO was implemented using the Optuna library in Python, with the Tree-structured Parzen Estimator (TPE) serving as the acquisition function to guide the hyperparameter search. The objective function was defined as the average validation loss from k-fold cross-validation, which enhanced generalization and reduced sensitivity to any single data split. For each sampled hyperparameter configuration, the BNN was trained and evaluated across $k=5$ folds, and the mean validation loss was returned to guide the BO search. A predefined search space was established for each tunable hyperparameter, including the number of neurons in each hidden layer, learning rate, and training steps (epochs). The optimization process iteratively updated a surrogate probabilistic model to identify promising regions of the hyperparameter space. Table 3 presents the option details and the final optimized hyperparameter values used in the BNN architecture.

The proposed BNN model as shown in Fig. 7, comprises an input layer with 14 parameters, connected to a first hidden layer with 68 neurons, followed by a second hidden layer with 150 neurons. Rectified Linear Unit (ReLU) activations are applied after each hidden layer to introduce non-linearity. The final output layer maps the 150-dimensional latent representation to 4 output logits, which are passed to a categorical likelihood function for multi-class classification. Training was performed using full-batch optimization (i.e., the entire dataset per gradient step) within the stochastic variational inference function, with the ELBO as the optimization objective.

The final model was retrained using these optimal hyperparameters, and the evolution of the ELBO loss over training iterations is shown in Fig. 8. As seen in Fig. 8, both training and validation losses exhibit a consistent downward trend, with the gap between them remaining narrow, indicating stable convergence and limited overfitting.

Table 3 Option details and tuned hyperparameters of BNN model

Hyperparameters	Option details	Tuned BNN hyperparameters
Neurons in 1st hidden layer	64–256	68
Neurons in 2nd hidden layer	64–256	150
Training steps or Epochs	5000–50000	20000
Learning rate	$10^{-4} - 10^{-2}$	0.00987

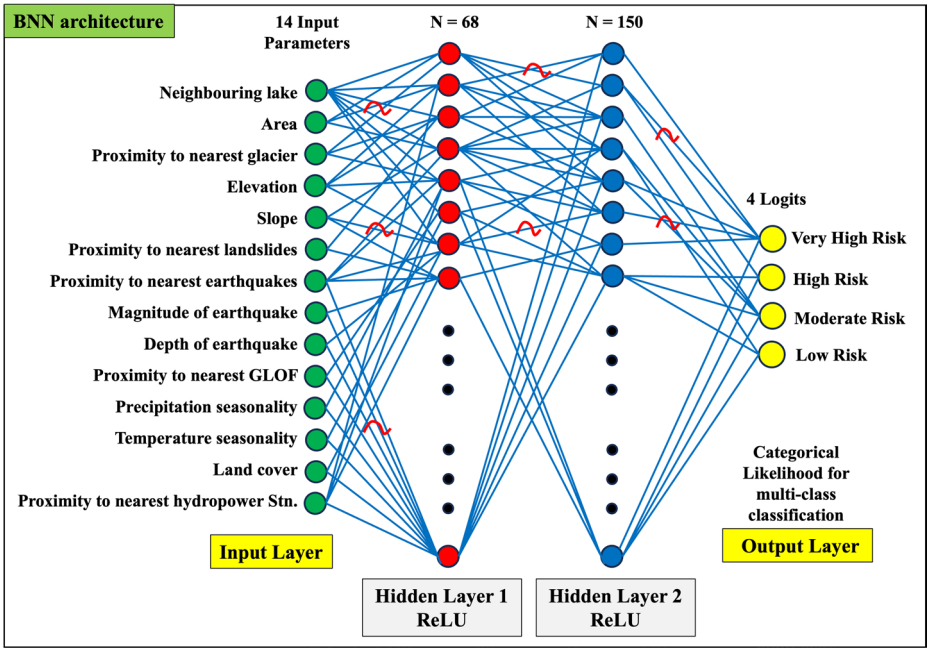


Fig. 7 Graphical representation of the proposed BNN model (N = number of neurons)

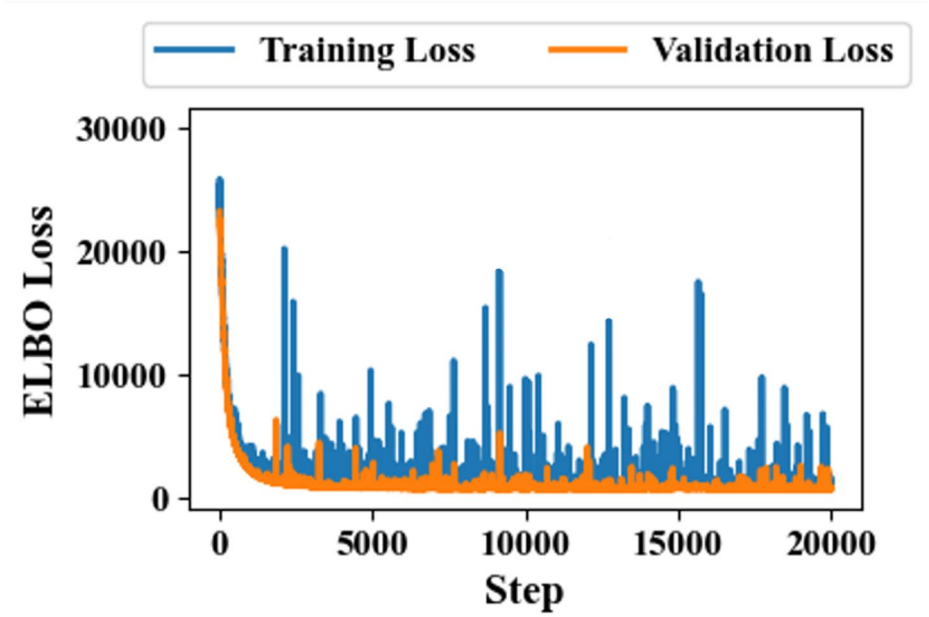


Fig. 8 Evolution of ELBO loss over training steps for the optimal BNN configuration (hidden layers: 68 and 150 neurons, learning rate: 0.00987, steps: 20,000). Both training and validation losses converge smoothly, suggesting stable learning dynamics

4.3 Accuracy and uncertainty estimate of GLOF risk predictions

The predictions of the trained BNN were generated using a Maximum A Posteriori (MAP) estimation approach. Specifically, 100 Monte Carlo forward passes were drawn from the predictive posterior distribution, and the class corresponding to the highest mean predicted logit across these samples was selected as the final prediction. This provides a computationally efficient and practical means of deriving point estimates from the probabilistic model.

Based on these MAP-derived predictions, classification performance was assessed using a confusion matrix, as shown in Fig. 9a.

The matrix shown in the Fig. 7a strong diagonal dominance, indicating a high level of agreement between the predicted (through BNN) and reference class labels derived from FCM clustering. Minor misclassifications were observed, mostly between adjacent GLOF risk categories such as 'High' and 'Very High', which reflects the gradual transitions in risk severity. Out of the 229 glacial lakes in the validation set (20% of the full dataset of 1144 lakes), the BNN model accurately classified 66 low-risk, 74 moderate-risk, 56 high-risk, and 18 very high-risk lakes, resulting in an overall classification accuracy of 93.4% in the validation set.

To further assess the model's probabilistic performance, receiver operating characteristic (ROC) curves were plotted for each risk class, as shown in Fig. 9b. The ROC curve plots the true positive rate against the false positive rate across varying classification thresholds, providing insight into the model's ability to distinguish between classes based on predicted probability scores.

The model achieved an Area Under the Curve (AUC) of 1.00 for the 'Low', 'High', and 'Very High' risk classes, indicating perfect discrimination between positive and negative samples for these classes. The 'Moderate' class also exhibited excellent performance with an AUC of 0.99, suggesting a near-perfect ranking of probabilities even in the presence of a few misclassifications. The micro-averaged AUC, which aggregates the performance across all classes, was also 1.00, underscoring the model's strong overall probabilistic performance.

Despite minor label-wise misclassifications noted in the confusion matrix, the high AUC values indicate the model's capacity to assign higher probability scores to the correct class

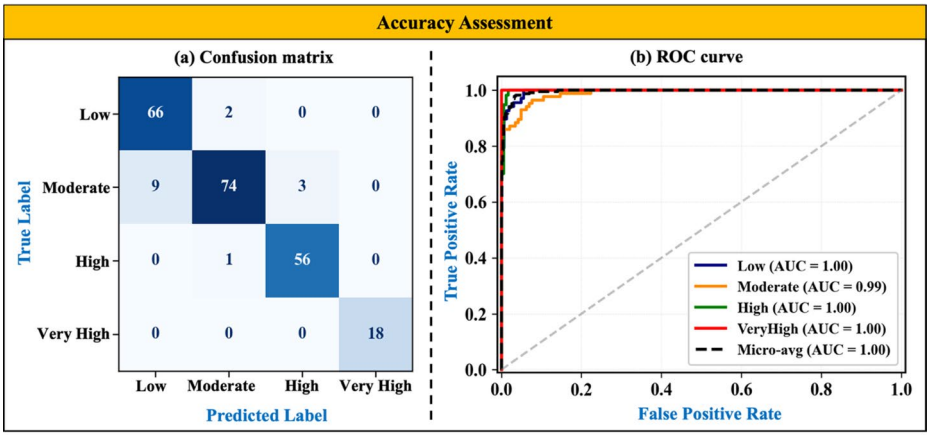


Fig. 9 Accuracy assessment of the BNN model on the validation set using the confusion matrix (**a**) and ROC curves with AUC scores (**b**)

consistently. Thus, ROC-AUC presents the ranking performance rather than hard classification accuracy. From Fig. 9, it is confirmed that the BNN model not only delivers high predictive accuracy but also produces well-calibrated probability distributions, supporting its application in GLOF risk classification and decision-support systems.

Figure 10 presents the spatial occurrence of predicted outcomes of the BNN model for the validation set, presenting very high-risk, high-risk, moderate risk and low-risk lakes. It also presents the potentially dangerous glacial lakes such as Imja Tsho, Tsho Rolpa, Barun Tsho, Luggye Tsho and South Lhonak lakes with their predicted risk levels.

The spatial distribution of 'high' and 'very high' GLOF risk glacial lakes are found to be primarily of large area, located at higher elevation with majorly rangelands and flooded vegetation in the neighbourhood. Additionally, these lakes are characterized by intense precipitation, and relatively higher magnitude of earthquakes. The glacial lakes farther from GLOFs, earthquakes, landslides with smaller area, mild slope and sparse connectivity to hydropower plants are found to have 'moderate' to 'low' GLOF risk. These findings are consistent with those reported in earlier research. For instance, studies like Sattar et al. (2021) and Kaushik et al. (2024) also documented that glacial lakes such as Khanchung and South

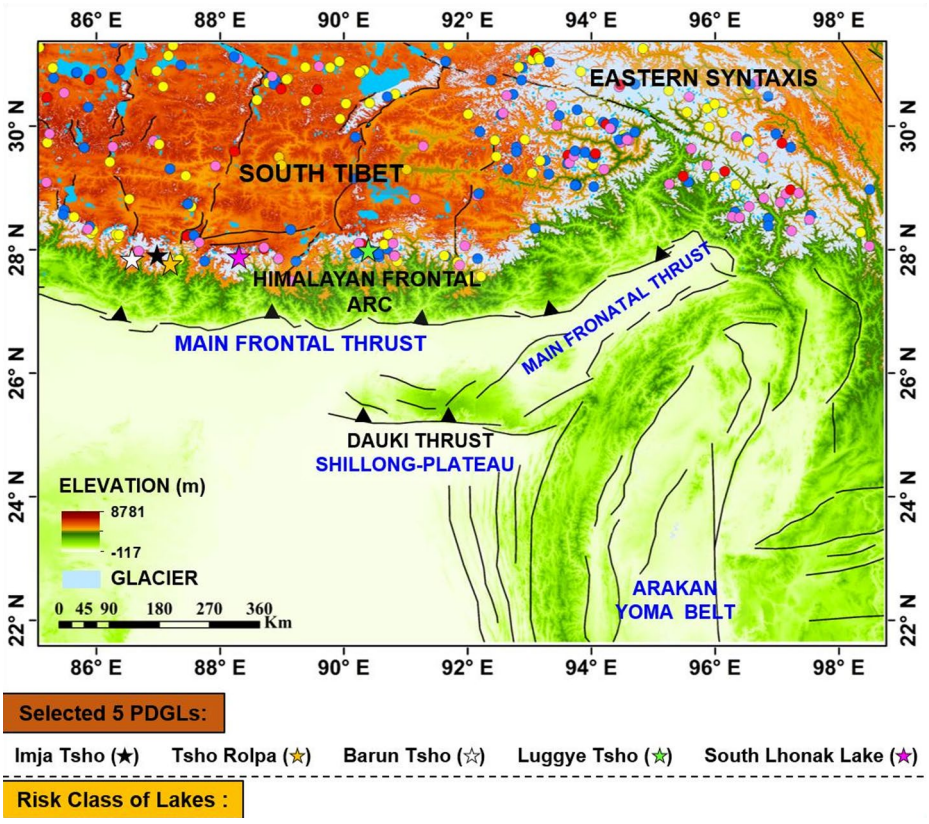


Fig. 10 Spatial distribution of glacial lakes classified by BNN into four GLOF risk levels across the Eastern Himalaya. Each point represents a lake location, color-coded by predicted risk category: Low (yellow), Moderate (blue), High (pink), and Very High (red)

Lhonak in Sikkim are particularly susceptible to GLOF events, influenced by factors like glacier retreat, precipitation variability, and seismic activity.

Figure 11 shows the spatial distribution of predictive uncertainty for glacial lakes in the validation set, as estimated by the BNN. Each lake is plotted by its geographic coordinates, with color intensity representing the variance in predicted class probabilities. Most lakes appear in blue shades, indicating low uncertainty and high model confidence, while a few scattered locations in orange to red represent relatively higher uncertainty. Out of the total 229 lakes, 142 (62%) exhibit a variance below 0.05, 59 (25.76%) fall within the range of 0.05 to 0.1, 25 (10.91%) lie between 0.1 and 0.15, and 3 (1.31%) show variance between 0.15 and 0.165. Overall, all predictions lie within a narrow uncertainty range, with the maximum variance not exceeding 0.165, indicating consistent classification performance across the study region. Further examination of the input data reveals that lakes with elevated uncertainty are typically very small in size, located at lower elevations, and situated near past GLOF and landslide sites. Moreover, such lakes may have recently formed or undergone morphological alterations following past failure events, further complicating their identification. The limited representation of these lake types in the training dataset likely contributes to the observed uncertainty.

Thus, the proposed approach contributes to strengthening early warning systems and adaptive risk management, supporting global initiatives under SDG 13 (Climate Action),

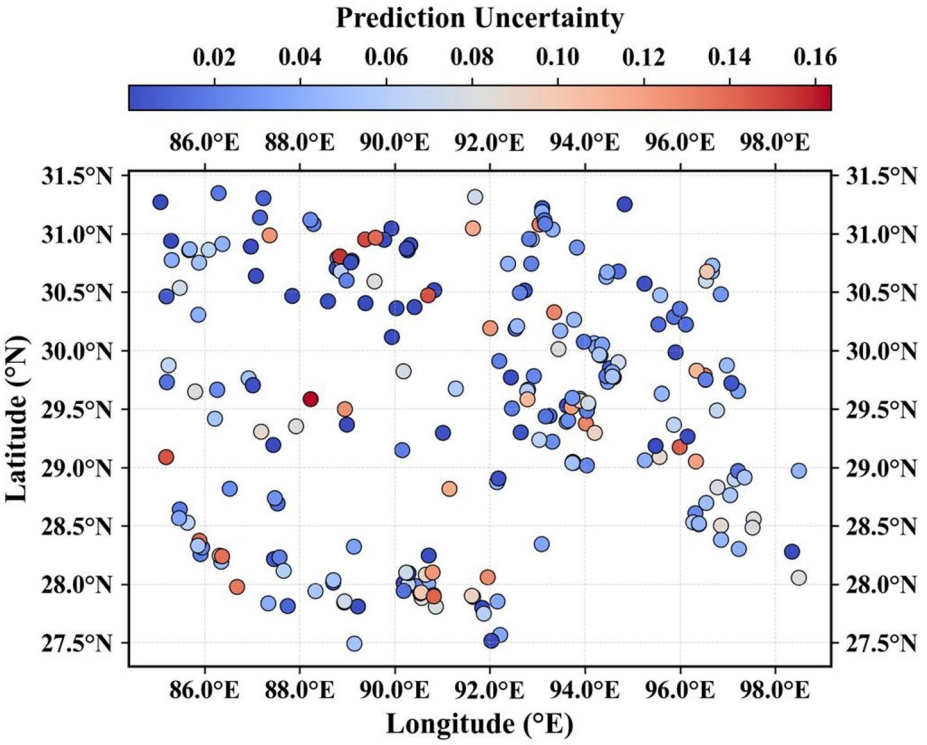


Fig. 11 Prediction uncertainty of the BNN model for glacial lakes in the validation set across the Eastern Himalayas

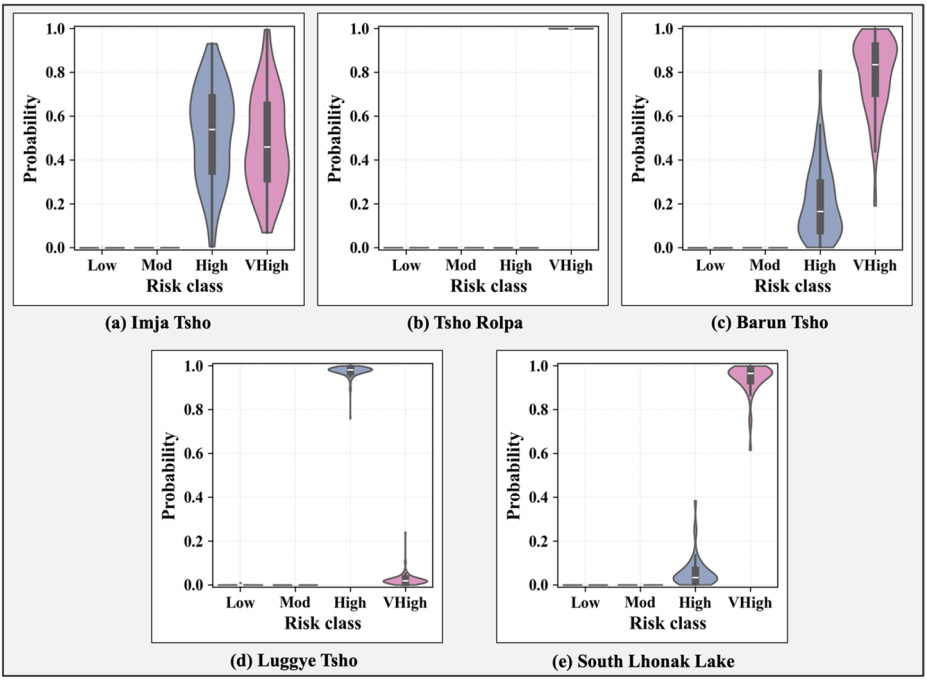


Fig. 12 Violin plots showing probabilistic GLOF risk classifications for five critical glacial lakes (**a** Imja Tsho, **b** Tsho Rolpa, **c** Barun Tsho, **d** Luggye Tsho, and **e** South Lhonak) using the BNN model

SDG 11 (Sustainable Cities and Communities), and SDG 15 (Life on Land) aimed at protecting vulnerable mountain regions from the impacts of a changing climate.

4.4 Probabilistic risk predictions of selected PDGLs

After developing the BNN model, it was applied to estimate the probabilistic GLOF risk levels for five glacial lakes identified as highly critical due to their historical evolution, geomorphological instability, and proximity to downstream settlements. These include Imja Tsho, Tsho Rolpa, Barun Tsho, Luggye Tsho, and South Lhonak Lake. The model output provides a probability distribution over four predefined risk classes low, moderate, high, and very high for each lake. To interpret these results, violin plots were generated for each lake (Fig. 12), where the width of each violin represents the density of the predicted probabilities, and the overlaid boxplot illustrates the interquartile range and median value within each class.

As shown in Fig. 9a, Imja Tsho (area = 1.55 km^2), located in Nepal, exhibits a nearly bimodal probability distribution between the "High" (0.519) and "Very High" (0.48) risk categories. This split reflects the model's uncertainty and indicates a transitional stage from high to very high risk. The total prediction uncertainty is medium at 0.0933. The elevated risk is primarily due to its proximity to glaciers and its location in a seismically active zone, necessitating annual monitoring.

Figure 12b presents Tsho Rolpa (area = 1.59 km^2), also situated in Nepal, as a clearly "Very High" risk lake, with a probability of 0.999. The distribution is sharply concentrated, with negligible support for other risk categories, and an extremely low uncertainty of 6.922×10^{-12}. Given its location near glaciers, the presence of the Khimti hydropower plant (60 MW) downstream, active earthquakes (M5.4), the Chubung glacial lake in the neighbourhood, and the potential destruction of trekking routes in the event of an outburst, the lake requires quarterly monitoring.

In Fig. 12c, Barun Tsho (area = 1.74 km^2), located in Nepal, is predominantly classified as "Very High" risk (probability = 0.788), with secondary support in the "High" category (probability = 0.211). The total prediction uncertainty is 0.0564. This classification is consistent with risk factors such as the presence of the Arun Hydropower Project (1061 MW) downstream, earthquakes (M5.6), and the threat of cascading outbursts due to the nearby Upper Barun Lake (Seto Pokhari). These factors highlight the need for regular monitoring.

Figure 12d illustrates Luggye Tsho (area = 1.57 km^2), located in the Lunana region of Bhutan, as primarily "High" risk with a dominant probability of 0.975. The distribution is tightly clustered with only a minimal tail into the "Very High" class, indicating strong model confidence and a low total uncertainty of 0.0015. The lake's risk is linked to the presence of neighbouring lakes, such as Druk Chung Lake, which could trigger cascading effects and the downstream presence of the PHPA I & II hydropower projects along the Punatsangchhu River. Also, the neighbouring lakes have a history of outburst floods, evidencing the need for annual monitoring.

Finally, Fig. 12e classifies South Lhonak Lake (area = 1.69 km^2), located in Sikkim, India, as a "Very High" risk with a probability of 0.941. The distribution is sharply peaked with a small tail extending into the "High" category (0.058), resulting in a low uncertainty of 0.0099. The lake's very high risk is associated with its setting in the Teesta River basin, a region prone to intense rainfall, frequent landslides, pronounced effects of climate change, and the occurrence of three hydropower plants downstream, necessitating quarterly monitoring.

In the literature, various studies provide different risk levels to the lakes. For instance, Bajracharya and Mool (2009), and Budhathoki et al. (2010) predicted Imja Tsho, Tsho Rolpa and Barun Tsho of Nepal as high-risk lakes. Further, a study by Rounce et al. (2016) predicted Imja Tsho as moderate risk, Tsho Rolpa as high risk, and Barun Tsho as very high risk lake. Next, studies like Sattar et al. (2021), Sharma et al. (2018), Wangchuk and Tsubaki (2024), and Komori et al. (2012), to mention a few examples of evidence that Luggye Tsho and South Lhonak lakes are potentially dangerous and need a risk assessment. Importantly, the state of the lake changes with time, owing to climate change, varying glacier-lake dynamics, topography and downstream exposure. Thus, the predictions of the present study, like Imja Tsho and Luggye Tsho as high-risk lakes, and Tsho Rolpa, Barun Tsho and South Lhonak Lake as very high-risk lakes, are well agreed with the published literature.

5 Conclusions

Despite significant progress in flood risk modelling, a persistent deficiency in methodological consistency remains a barrier to the effective translation of research findings into policy and practical risk management. This challenge is especially pronounced in the Himalayas,

where prompt and reliable information is crucial for hazard preparedness. Previous efforts have largely relied on additive linear weighting methods to synthesize multiple criteria; while intuitive, these heuristic approaches often lack theoretical rigor and fail to adequately account for the inherent uncertainties and nonlinearities of glacial lake systems. In response to these limitations, our study presents a unified, uncertainty-aware probabilistic framework that represents a significant methodological advance in GLOF risk assessment. By integrating 14 critical parameters across geomorphological, topographical, meteorological, and anthropogenic domains, and combining expert-driven AHP with the objectivity of Shannon Entropy, we establish a hybrid multi-criteria decision-making model rooted in both domain knowledge and data variability. The subsequent use of the TOPSIS method and Fuzzy C-Means clustering yields soft-labeled risk classes, enabling a nuanced classification of glacial lake hazards. The introduction of a BNN–trained on these risk labels–further enhances this framework by modelling nonlinear relationships while providing probabilistic predictions enriched with explicit uncertainty quantification. The results, drawn from an assessment of 229 glacial lakes, reveal a high overall predictive accuracy (93.4%), with 87.77% of lakes demonstrating low prediction variance (below 0.1), underscoring the model's reliability. The framework successfully classifies Imja Tsho and Luggye Tsho as high-risk, and Tsho Rolpa, Barun Tsho, and South Lhonak Lake as very high-risk—findings that resonate with prior field-based observations while offering enhanced statistical confidence. In sum, this study contributes a scalable and generalizable approach to GLOF risk evaluation, addressing a critical gap in the current hazard modelling landscape. It supports the development of more resilient early warning systems and adaptive planning strategies, advancing global efforts aligned with SDG 13 (Climate Action), SDG 11 (Sustainable Cities and Communities), and SDG 15 (Life on Land) to safeguard vulnerable mountain communities in a warming world.

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